Design, Development and Validation of a Ground Control Station for Aerial Manipulators

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Abstract. In the last years, aerial manipulators, UAVs (Unmanned Aerial Vehicles) with robotic manipulators, have been used for several applications such as inspection tasks, systems monitoring or transportation of loads. In order to supervise the missions, a GCS (Ground Control Station) is used. However, the GCS of the commercial autopilots are just designed to monitor the state of the aerial platform, forcing the use of other software to monitor the state of the robotic manipulator. This paper presents the design, development and validation of a Ground Control Station for aerial manipulators. The GCS developed in this work allows the integration of the UAV state and manipulator state in the same GUI (Graphical User Interface). In order to plan missions, it also allows the definition of waypoints in a simple way. In addition, the GCS integrates a slave mode for teleoperation of the aerial manipulator, using a ground manipulator and a video streaming to facilitate the task of the ground operator. All the functionalities of the GCS are validated through experiments. This GCS will be useful to perform aerial manipulation mission, allowing the monitoring of the complete aerial manipulator in only one software.

Keywords: Aerial Manipulators · Ground Control Station · Multirotors

1 Introduction

In recent years, the fields of application of aerial manipulators have expanded, allowing the manipulation of objects in flight [1][2] and the interaction with the environment [3]. Some application examples are the inspection of oil and gas refineries, chemical plants [4], power lines [5], or civil infrastructures [6,7]. In order to monitor these missions, a GCS (Ground Control Station) is necessary. Ground Control Stations have an important role in the aerial manipulation, as they are used for planning missions, monitoring the status of the aerial platform or establishing communication links. Some good examples of GCS are *QGroundControl* [8], *Mission Planner* [9] or *MAVProxy* [10].

However, these GCS are primarily intended for the flight of aerial vehicles, without taking into account the robotic manipulator. In addition, most of them are designed for outdoor flight, using GPS. This makes it difficult to integrate an indoor positioning system such as OptiTrack. Furthermore, waypoints are

defined in geographical coordinates, longitude and latitude, rather than in local axes, referenced to the centre of the OptiTrack system, for example.

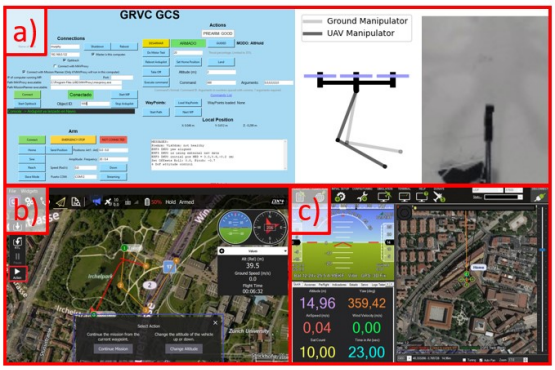


Fig. 1: GCS examples: a) GRVC GCS b) QGround-Control c) Mission Planner.

These disadvantages can be overcome by incorporating a second on-board computer to receive the OptiTrack position and waypoints, typically via ROS, and transmit these messages to the autopilot. An on-board computer can relieve the autopilot of complex calculations, but it increases the weight, consumption and cost of the aerial platform. Although some GCSs allow the incorporation of different actuators in the multirotor, such as a gripper, a camera or a parachute release mechanism; they are not designed for a multirotor with a robotic manipulator.

So, the main contribution of this paper is the design, development and validation of a Ground Control Station that incorporates the state of the robotic arm, as Figure 1a shows. Furthermore, the GCS developed can incorporate the indoor positioning system OptiTrack in a simple way, as well as the waypoint definition on local axes. Finally, the GCS incorporates different manipulator control modes, such as manual mode, predefined movements or slave mode with a ground manipulator.

The paper is structured as follows: Section 2 introduces the main feautures of the GCS designed. Section 3 presents the software structure of the Ground Control Station. Section 4 introduces the aerial platform used for experiments. Section 5 covers the validation of the GCS through indoor experiments using the OptiTrack system. Finally, Section 6 summarizes the conclusions and proposes future lines of work.

2 Features of the Ground Control Station

This section presents the GUI (Graphical User Interface) of the GCS designed. Figure 2 presents the GUI of the GCS, divided in five different regions. The utilities of each region are presented below.

Region A: Connections. In this region, the connections with the autopilot are configured. As Figure 2a shows, the GCS allows the use of OptiTrack system, as well as the use of the others GCS (MAVProxy or Mission Planner) at the same time. This feature has been incorporated in order to reuse the characteristics of other GCS, such as the GUI for the parameters of Mission Planner. In addition, this region shows the status of the autopilot connection, which is continuously checked.

Region B: Robotic Arm Control. This region integrates the state of the robotic arm, as Figure 2b shows. The connection with the robotic arm software is established in this region. In this case, the robotic arm is a 2DoF (Degree of Freedom) manipulator. However, the code can be easily changed for the modification of the arm configuration. The robotic arm can be controlled in manual mode, by sending the joint coordinates. Furthermore, two predefined movements are available: a sine movement in the shoulder articulation, while the elbow articulation is blocked, and a reach movement, where the arm extends and retracts to the maximum. Finally, the robotic arm can be controlled using a master arm in ground. In this mode, the GCS represents both the slave manipulator (UAV manipulator) and the master manipulator (ground manipulator), as Figure 3a shows. This region also allows the activation of a video streaming platform.

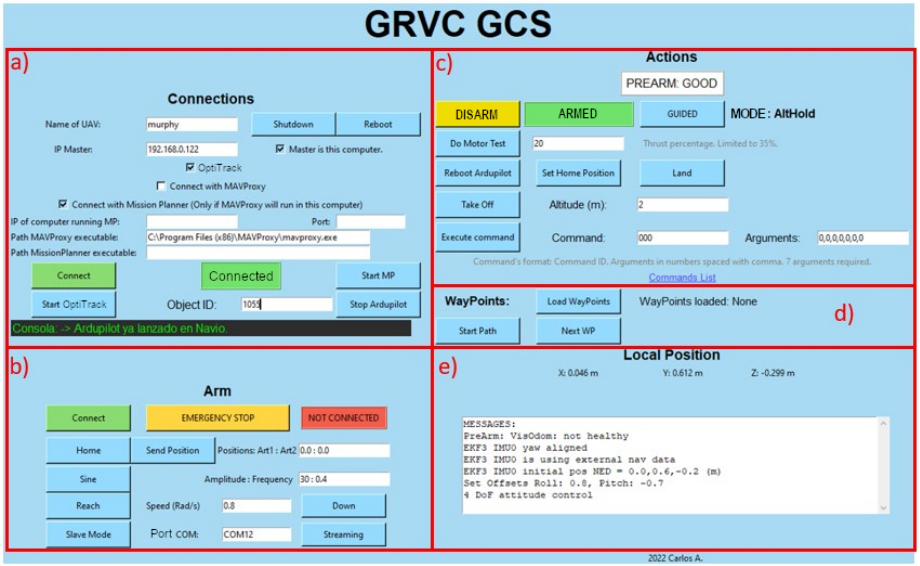
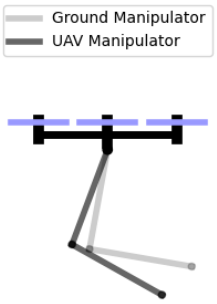


Fig. 2: GUI of the GRVC GCS.

Region C: Actions of the Aerial Platform. This region allows some predefined actions, as Figure 2c shows. Some of them are: arming/disarming the aerial platform, testing the motors, rebooting the autopilot or landing/taking off

the platform. If some MAVLink command not defined in the GUI needs to be sent, it can be done by using the command ID as well as the necessary arguments.



(a) Arms in the slave mode.

```
#Waypoints definition

#Format: x,y,z,xdot,ydot,zdot,yaw,yawdot
#Types of WP:
# -Absolute: in the Optitrack frame.
# -Relative: in the UAV frame.

# Use # to comment

#Example:
Absolute
0,0,3,0,0,0,0,0
Relative
1,0,0,0,0,0,0,0
```

(b) Waypoint definition example.

Fig. 3: Some features of the Ground Control Station

Region D: Waypoints definition. This region allows the definition of the waypoints of a mission. The waypoints are defined in a *txt* file, using the format presented in Figure 3b. Once that the waypoints have been loaded, the mission can start.

Region E: Messages. Finally, this region shows the messages that the autopilot send to the GCS. In addition, it shows the current position of the platform.

3 Implementation of the Ground Control Station

The Ground Control Station has been developed completely in Python, using *Tkinter* for the design of the GUI (Graphical User Interface), and *Pymavlink* for the implementation of the MAVLink protocol.

3.1 Threads

The software consists of five main threads, plus auxiliary threads, running in parallel. Figure 4 summarizes the main threads, and their inputs and outputs.

1) GUI thread. It is responsible for updating the content on the screen, as well as responding to button presses.

2) MAVLink reception thread. This thread receives the messages sent to the GCS by the aerial manipulator using MAVLink protocol. It also processes them and generates the corresponding actions. It continuously checks the connection status, displaying on screen if the connection with the aerial platform has been lost.

3) OptiTrack thread. It receives the messages sent by Motive, the OptiTrack software, converts them into MAVLink message and sends them to the aerial manipulator.

4) Manipulator thread. This thread is in charge of several tasks related with the manipulator of the aerial platform. It continuously checks the status of the connection with the manipulator of the UAV, displaying a message on the screen if the connection is lost. It updates the GUI with the values of the arm joints. In the *Slave Mode*, it reads the state of the master arm, and sends these coordinates as references to the arm of the aerial platform.

5) Streaming thread. This thread receives the images sent by the UAV and displays them on the screen.

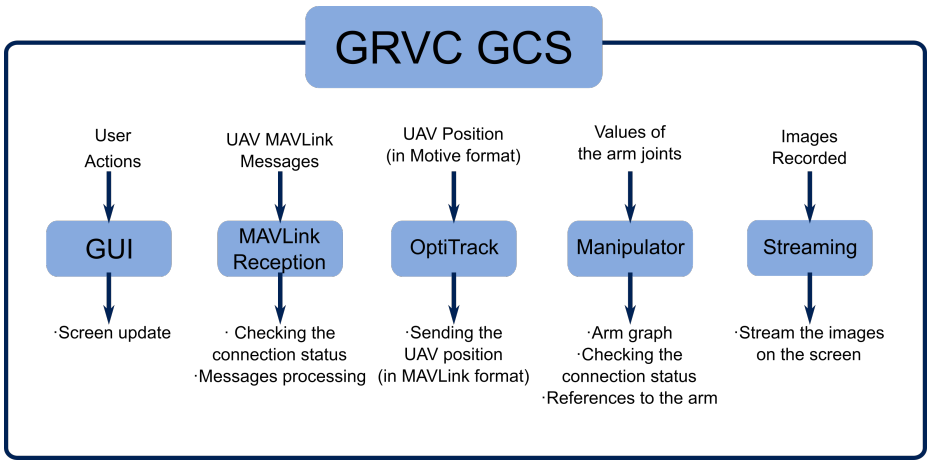


Fig. 4: Threads of the GCS software and their functions.

3.2 Communications

The GCS establishes a connection between the different devices. This connection is either wireless, as with the aerial manipulator, or by cable if the other device is close to the ground computer, like with the ground manipulator used in the slave mode. Figure 5 shows a general scheme of the connections to the GCS. The connection between the autopilot and the GRVC GCS is established using the MAVLink protocol. MAVLink has been configured to operate over UDP protocol. In this way, fast data transmission is achieved, although with lower reliability than the TCP protocol. In the same way, the connection between the GRVC GCS and other GCSs is established via MAVLink over UDP.

The UAV's manipulator is connected to the UAV's Raspberry Pi via the serial port. In this way, the GRVC GCS communicates wirelessly with the manipulator via the Raspberry Pi. There are two connections to the GRVC GCS. One via UDP protocol, by which references of movement and manipulator status are sent, and another connection by which an emergency stop would be sent, via TCP protocol to increase reliability. In the same way, the camera is connected via USB to the Raspberry Pi, which sends the image to the GRVC GCS using the UDP protocol.

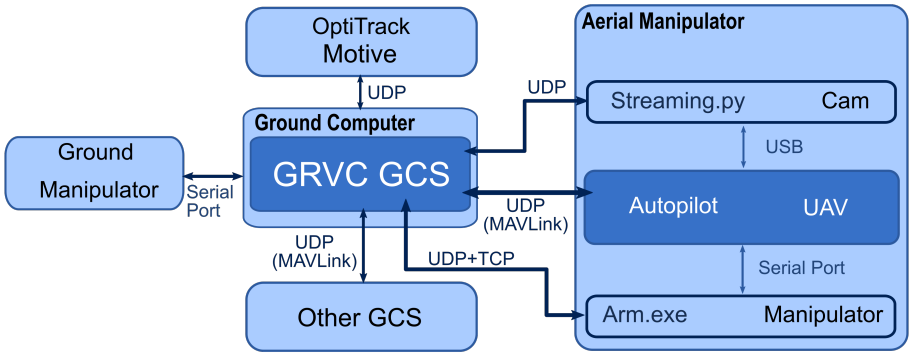


Fig. 5: Connection diagram between the GRVC GCS and other devices

In addition, the OptiTrack system publishes the position of the UAV via UDP protocol. The GCS establishes a UDP connection to Motive, the OptiTrack software, and receives the messages in order to send them to the autopilot. Finally, the ground manipulator, used during slave mode, is connected to the GRVC GCS via serial port.

4 System for the Experimental Validation

For the experimental validation of the GCS in Section 5, an aerial manipulator, presented in Section 4.1 and a ground manipulator for the slave mode, presented in Section 4.2, have been designed and built.

4.1 Aerial Manipulator

The aerial manipulator used for the experimental validation of the GCS is composed of an hexarotor as aerial platform and of a 2DoF manipulator. Figure 6 shows all the components of the aerial manipulator.

The aerial platform is composed of a DJI F550 frame, which includes the DJI 2312E rotors as propulsive system. It also includes the DJI 9x4.5 propellers and the XRotor 40A ESCs. The autopilot selected for the aerial platform is

Ardupilot, running in a Raspberry Pi 4 connected to the Emlid's sensors shield Navio2. To stream the video, the aerial platform includes the Logitech C920 HD PRO camera. The frame has been modified using 3D printed pieces in PLA in order to increase the space in the avionics bay. All is powered by a 4S LiPo battery.

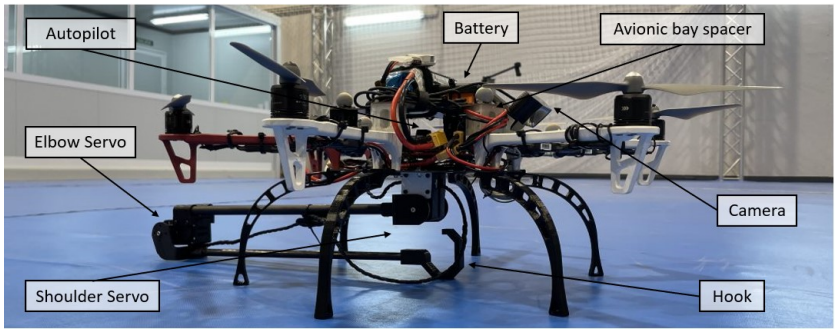


Fig. 6: Parts of the aerial platform used during the experimental validation.

The aerial platform is equipped with a 2DoF manipulator. The 2DoF arm is composed of a Herkulex-DRS-0602 for the shoulder joint, and a Herkulex-DRS-0402 for the elbow joint. Link 1 and link 2 are composed of carbon fiber tubes. The assembly between the joints and the links are made using ABS 3D printed pieces. In order to manipulate objects, the manipulator is equipped with a hook at the end. The manipulator is powered by the 4S LiPo battery of the hexarotor, using a buck converter to reduce the supply voltage.

To control the aerial manipulator, a decoupled control scheme is used, i. e. the manipulator and the aerial platform are controlled as two separated systems. In Slave Mode, the angles of the ground arm are readed and sent directly to the UAV arm. Then, the internals controllers of the Herkulex servomotors take them as references to move the arm.

4.2 Manipulator for the Slave Mode

To validate the Slave Mode of the GRVC GCS, another 2DoF manipulator has been designed and built. This manipulator is similar to the manipulator of the aerial platform, since it has the same joints: an elbow joint and a shoulder joint. Likewise, the links are made with carbon fiber tubes, and the union between the links and the joints are made with ABS 3D printed pieces. However, as it is handled by the operator and does not have to exert any effort, the joints are equipped with less powerful servos. For both joints, an Herkulex-DRS-0102 has been selected. Furthermore, this arm has been built with a 3:4 scale with respect to the aerial manipulator arm. As Figure 7 shows, this master manipulator is installed in a ground structure to facilitate its operations.

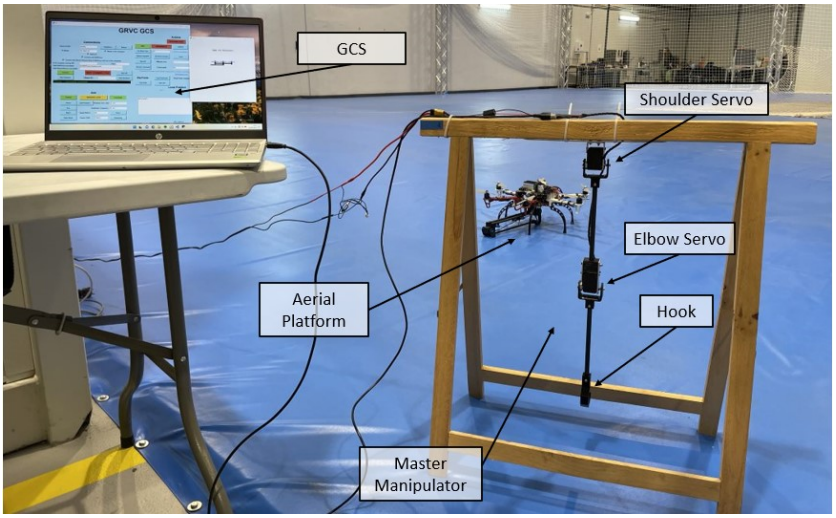


Fig. 7: Master 2DoF manipulator in the slave mode.

5 Experiments

The experimental validation of the GCS is done through three different experiments. First, the integration of OptiTrack system is tested by performing a defined mission with waypoints in Section 5.1. Then, the two predefined movements of the aerial manipulator are validated in Section 5.2. Finally, Section 5.3 validates the slave mode by a capture operation of an object.

All the experiments are carried out in the GRVC indoor facilities, which is equipped with the OptiTrack positioning system composed of 28 cameras. The videos of the experiments are available in [11].

5.1 Mission using waypoints

For validating the integration of the OptiTrack system within the GRVC GCS, as well as the waypoint sending system, a mission has been programmed using the waypoint structure presented in section 2.

Table 1: Coordinates of the trajectory points.

	Initial	WP1	WP2	WP3	WP4	WP5	End
X [m]	-4.8	-4.8	-2.8	-2.8	-4.8	-4.8	-4.8
Y [m]	3.0	3.0	3.0	1.0	1.0	3.0	3.0
Z [m]	0.0	2.0	2.0	2.0	2.0	2.0	0.0

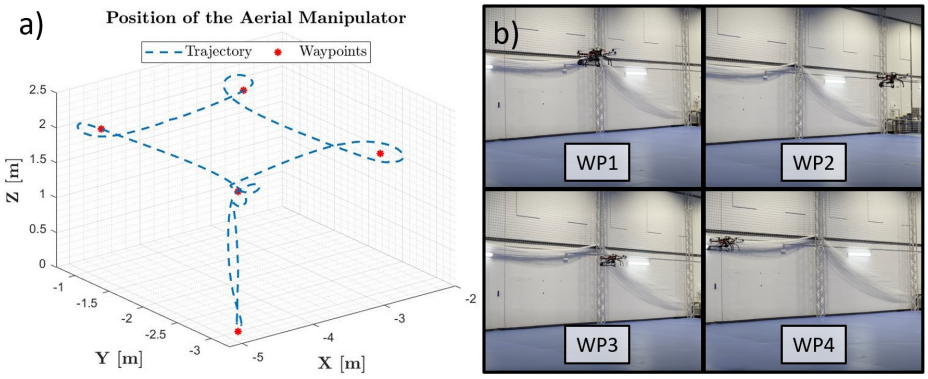


Fig. 8: Square trajectory: a)Trajectory of the UAV b)UAV during the mission.

The mission performed is a square trajectory. Table 1 shows the coordinates of the initial point, the five waypoints of the square and the end point. Figure 8a shows the trajectory of the aerial manipulator during the mission. In addition, some pictures of the aerial platform during the mission are added in Figure 8b. As the mission has been successfully completed, the integration of OptiTrack and the sending of waypoints in the GRVC GCS is validated.

5.2 Predefined movement of the aerial manipulator

After the validation of the integration of OptiTrack in the GRVC GCS, the predefined movements of the aerial manipulator are validated. These movements are presented in Section 2.

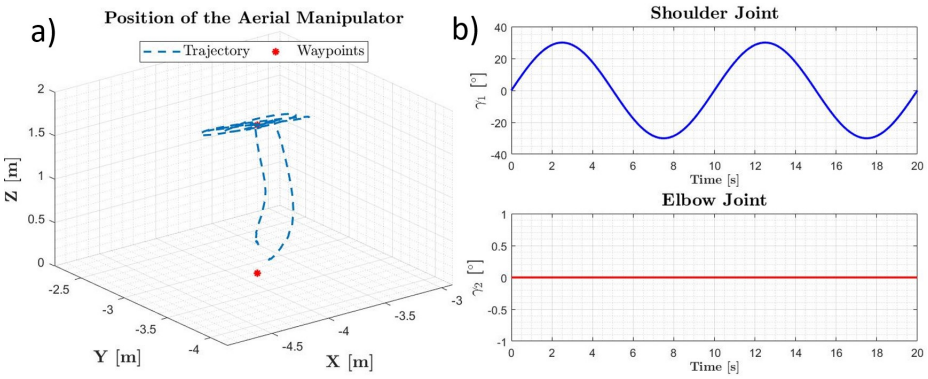


Fig. 9: Predefined movement A: a)Trajectory of the UAV b)Angles of the joints.

Both movements are validated in the same way: the UAV takes off autonomously, the aerial manipulator performs the movement to be validated and finally the UAV lands once the arm is retracted. Both movements are validated at low speeds, to avoid disturbing the aerial platform too much. Figure 9a and Figure 10a show the trajectories of the aerial platform during the experiment.

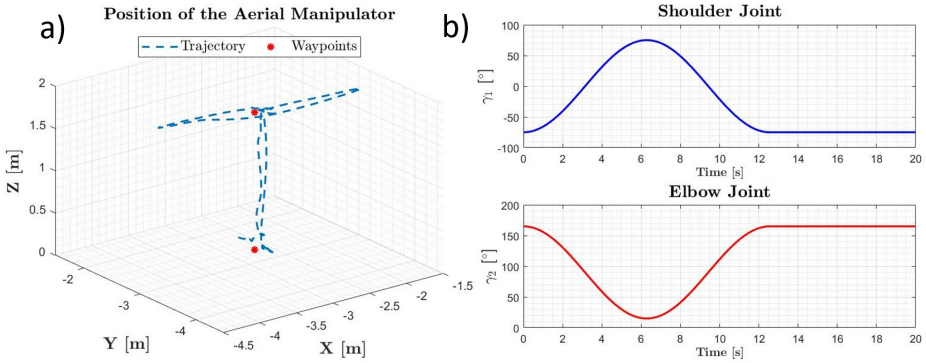


Fig. 10: Predefined movement B: a)Trajectory of the UAV b)Angles of the joints.

The deviations from the initial point are due to the movement of the manipulator, which disturbs the aerial platform. In addition, Figure 9b and Figure 10b show the values of the joints arm during the predefined movements.

5.3 Aerial Manipulation Mission with the Slave Mode

Finally, the validation of the video streaming and the slave mode is performed with an aerial manipulation mission.

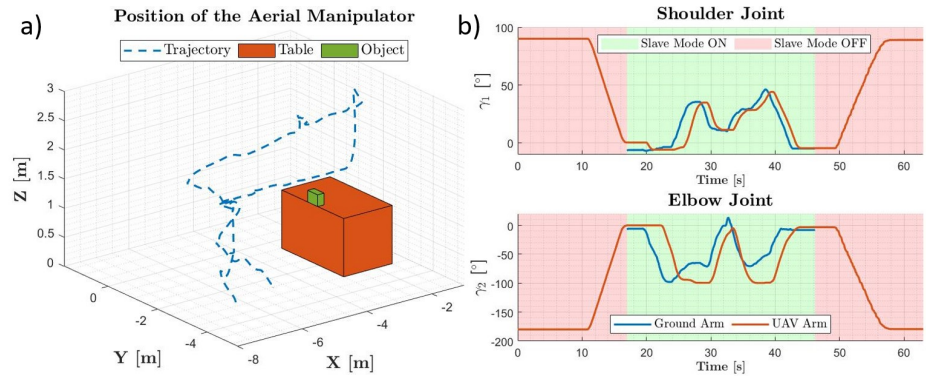


Fig. 11: Mission in slave mode: a)Trajectory of the UAV b)Angles of the joints.

The mission consists of picking up an object on a table with the hook, teleoperating the aerial manipulator in slave mode with the ground manipulator. In addition, the ground operator will use the video streaming to perform the capture of the object. Figure 11a shows the trajectory of the aerial manipulator during the mission. In addition, Figure 11b shows the values of the joints during the mission. Furthermore, some pictures of the mission are represented in Figure 12. After these three experiments, the GRVC GCS is completely validated.

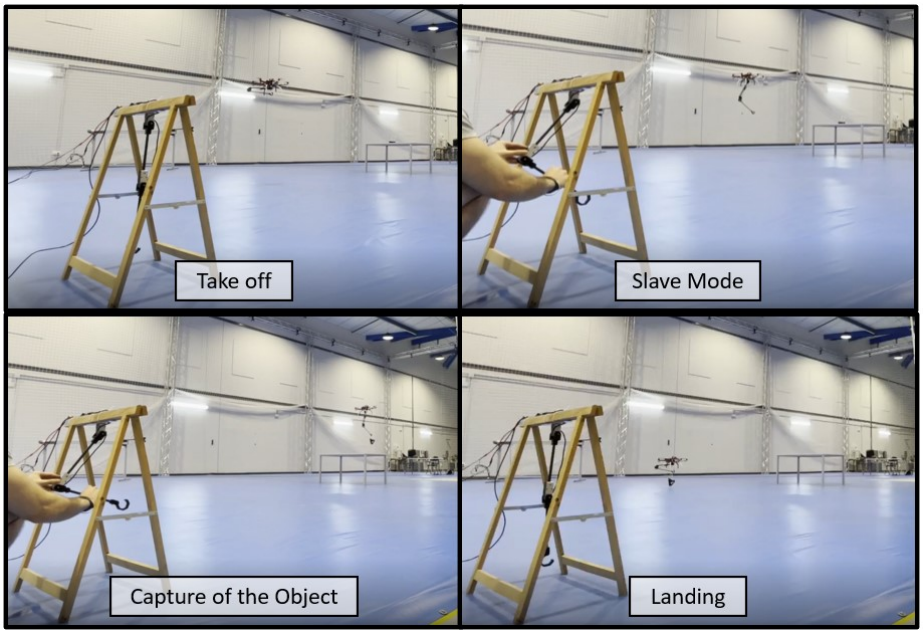


Fig. 12: Some instants of the mission in slave mode.

6 Conclusion

This paper has presented the development and validation of a Ground Control Station for Aerial Manipulators. The objective of this GCS is to integrate the information of the manipulator and the aerial platform in only one GUI. The paper has presented a functional design that allows the integration of the OptiTrack positioning system, the sending of messages to the autopilot and the simple definition of waypoints. The solution presented also allows the control of the aerial manipulator in different modes (sending joints coordinates, with predefined movements or slave mode) and the video streaming to facilitate the manipulation tasks. All the functionalities of the GRVC GCS have been tested in indoor tests with an aerial manipulator.

It is proposed as future work the integration of the totalstation positioning system. In this way, the GCS could be used to perfom outdoors experiments using a positioning system more accurate than GPS.

Acknowledgment

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